

TECHNICAL UNIVERSITY OF DENMARK

12102 ENVIRONMENTAL PROCESSES

Grand Exercise

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1 Summary

The purpose of this report is to assess the effects of a point source batch discharge of 10 kg DCB, and its degradation product BAM, in a lake with a nearby ground water well, used for drinking water.

Technical specifications were defined, though some assumptions were required to complete the analysis. To ensure reliability of results regarding safety- and health protocols a worst-case-scenario approach was applied.

Results found no critical toxicological effects for maximum concentrations of DCB at 0.258 mg/L and BAM at 20 $\mu g/L$. Sand filtration was found ineffective, while active carbon filtration is viable requiring periodic replacement every 12.2 years to stay below 0.1 $\mu g/L$ concentration of BAM. Aeration is necessary to manage calcium carbonate precipitation and reduce iron levels below 200 $\mu g/L$, according to danish regulatory. Suitable reactor designs, including CMFR-in-series and PFR, were evaluated to achieve this. An alternative well was assessed and showed extreme water hardness at 37 $^{\circ}dH$, likely requiring water softening.

The report concludes that continued water extraction from current well is possible with activated carbon filtration and aeration to comply with the Danish drinking water BAM limit of 0.1 $\mu g/L$.

2 Introduction and aim

Based on increasing concentration levels of pesticides, our groundwater is at risk. A frequently found compound in danish groundwater is 2,6-dichlorbenzamide (BAM), which originates from its parent compound 2,6-dichlorbenzonitraile (DCB). Dichlorbenide has been banned in Denmark since 1997. In this exercise we will investigate an incident where a 100 kg bag of "Præfix", containing 10 kg of the pesticide DCB has been dumped into a lake by a farmer. The lake, today known as *Lake Gravel Pit* is located in a small town on Southern Funen. The situation is described by Figure 1 below, where each number, 1 through 7, indicates a new process step in the system.

To evaluate the effects of the pollution on the ecosystem, the concentration for both DCB and BAM will be determined to begin with. Then we consider the water being transported to a nearby water work, where the impact of the pollution will be evaluated. Afterwards treatment processes, such as aeration, activated carbon filters, and sand filters will be evaluated based upon calculations. Finally it will be determined if the pesticide-contaminated water is suitable for potable water production without exceeding limit values in regards to Danish legislation, or whether another, uncontaminated water well is required.

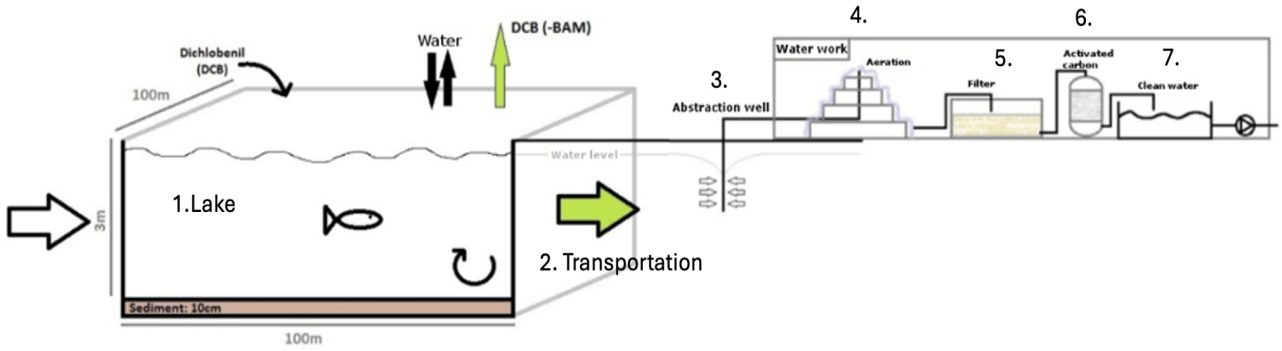


Figure 1: *Overview of the lake system, transportation of water, and subsequent water treatment processes.*

Some general data on Lake Gravel Pit has been advised, and is used for calculations throughout the exercise. It can be seen in Table 1 below.

Table 1: *General data on Lake Gravel Pit used in calculations*

Side lengths of lake	100 m
Depth of lake	3 m
Darcy velocity, through one side	20 m/year
Water work production	300,000 m^3 water per year

3 Evaluation & Data Treatment

To understand the effects and extent of the pesticide pollution on the surrounding environment the water quality and the different treatment procceses at local water works are examined. In the following section the respective concentration of DCB and BAM in the lake are determined, along with an evaluation of whether the levels are toxic for the living organisms in the lake. According to Danish legislation the limit value of a single pesticide is given as 0.1 $\mu g/L$ and the total limit value for all pesticides as 0.5 $\mu g/L$ (Baun, 2024). Therefore the extend of the pollution will be examined by looking at these specific limit values, if it causes problems for the local waterworks, and including possible need for additional treatment processes. Finally an alternative water well is examined, to understand if it would be suitable as a replacement for the existing water well supplying the water work.

3.1 Pollution and Toxicity of DCB and BAM in *Lake Gravel Pit*

At first we are situated in the lake, at point 1 in Figure 1, where the initial concentration of DCB has to be determined. Hence some basic assumptions are necessary. It can be assumed that the first 10 cm of the sediment at the bottom of the lake is in direct contact with the water and limitation due to transport can in this case be neglected, assuming the sediment is completely mixed with the water (Kaarsholm et al., 2024).

Firstly, the initial concentration of DCB is calculated in the lake using the following equation,

since linear sorption is assumed:

$$C_s = K_d \cdot C_w \iff K_d = \frac{C_s}{C_w} = f_{oc} \cdot K_{oc} \tag{1}$$

Where C_s refers to the concentration in the solid phase (mg/kg), K_d is the partition-coefficient (L/kg), and C_w being the concentration in the aqueous phase (mg/L), which is the value of of interest (Bjerg et al., 2023).

To calculate the K_d value, we first have to determine values of f_{oc} and K_{oc}. The organic carbon fraction is given as 2%, and K_{oc} is calculated using the $logK_{oc}$ and $logK_{ow}$ correlation. Since we are looking at lake sediments, equation *C* in Table 6.2 (Bjerg et al., 2023) is used for the values of a and b. From this we obtain the following: $logK_{oc} = 0.72 \cdot 2.7 + 0.49 = 2.434$.

From the previous values, the K_d value can be determined as following:

$$K_d = f_{oc} \cdot K_{oc} = 0.02 \cdot 10^{2.434} = 5.43L \cdot kg^{-1}$$

By using the law of mass balance for compounds, we are able to determine the value of C_w, by using the following expression to isolate for C_w and substituting with the previous equation for C_s.

$$M_t = C_w \cdot V_w + C_s \cdot M_s, \;\; C_s = K_d \cdot C_w \tag{2}$$

From this we obtain:

$$C_w = \frac{M_t}{V_w + K_d \cdot M_s} \tag{3}$$

M_s is the mass of solid matter in the total volume, $M_s = \rho_b \cdot V_w$ (Bjerg et al., 2023). The value can be determined by using the given bulk density from Appendix 2 (ρ_b) and by calculating the volume of sediment in the lake. Obtained values are inserted in *Equation* 2, from where the initial DCB-concentration in water, C_w is determined. (All detailed calculations can be found in Appendix 1, section 1A).

$$C_w = 2.584473 \cdot 10^{-7} \text{ kg /L}$$

3.1.1 Mass Balances for DCB and BAM

To model the different transport processes for DCB in the lake we set up a mass balance to understand which processes are the main contributors to the removal of DCB. In this situation we are located at the transportation step, equivalent to point two in Figure 1:

Inflow + Net Transformed = Outflow + Accumulated

As we only observe the scenario after the "Præfix" is dumped into the lake, we can set the inflow process in the equation equal to zero (Inflow $= Q_{in} \cdot C_{in}[mol \cdot s^{-1}] = 0$).

According to Darcy's law we know the following can be substituted in the outflow process in the equation later, $Q_{out} = U \cdot A_{out}$, and we set $C_w = C(t)$, where A_{out} is the cross-sectional area $[m^2]$ of one side of the lake, where the outflow flows through, and U is the Darcy Velocity $[m \cdot s^{-1}]$. The substitutions results in the following expression:

$$\text{Outflow} = Q_{out} \cdot C_w = U \cdot A_{out} \cdot C(t)\ [mol \cdot s^{-1}]$$

When applying the term of accumulation according to *Equation* 2, we can substitute $C_s = K_d \cdot C_w$:

$$\text{Accumulated} = \frac{d(C_w \cdot V_w)}{dt} + \frac{d(C_s)}{dt} \cdot M_s = \frac{d(C_w \cdot V_w)}{dt} + \frac{d(K_d \cdot C_w)}{dt} \cdot M_s$$

The transformation process is divided into two separate subprocesses: degradation and evaporation. The degradation process follows first order decay, and according to Table 5.A.1 (Nazaroff and Alvarez-Cohen, 2001) the reaction rate, r, is given by $r = -k_1 \cdot C$.

$$\text{Degraded} = r \cdot V = -k_1 \cdot C(t) \cdot V_w\ [mol \cdot s^{-1}]$$

The value of the first-order rate constant k_1 is determined from the given half life value $t_{1/2}$ for DCB in the lake, provided in the technical data (Appendix 2). Since the relationship is of first order kinetics, k_1 is calculated accordingly. Detailed calculation is provided in Appendix 1 section 1B.

In the evaporation process the net flux of DCB in the lake across the air-water interface has to be included in the derivation. To describe the transport rate, we use the expression of $J_{gl} = k_{gl}(C_s - C)$. As we are not considering a situation of equilibrium in this case, the value of C_s can be neglected (Nazaroff and Alvarez-Cohen, 2001) and the expression becomes as following:.

$$\text{Evaporation} = k_{gl} \cdot A_{surface} \cdot C(t)$$

The calculation for the value of the mass transfer coefficient k_{gl} is provided in Appendix 1, section 1B.

As we have obtained expressions for each term in the material balance, the final differential equation for DCB is expressed as:

$$\frac{d(C(t))}{dt} = \frac{-(U \cdot A_{out})\, C(t)\ -\ \left(k_{gl} \cdot A_{\text{surface}} \cdot C(t) + k_1 \cdot V_w \cdot C(t)\right)}{V_w + K_d \cdot M_s}$$

Correspondingly a mass balance is also set up for BAM to investigate the concentration over time for the pesticide. From calculating the fraction of water we obtain a value of $f_w = 99\%$ (Appendix 1, section 1C). We observe that the vast majority of the pesticide is in the water with only 1% being sorbed to the sediment, and therefore we can neglect sorption in this

situation. According to PubChem BAM has a very slow degradation and is extremely persistent, therefore degradation may also be neglected (PubChem, 2024). Lastly we consider the evaporation term by using Henry's law, which is an expression for the air-water distribution ratio. The Henry's law constant for BAM has a table value of $K_{H,BAM} = 1.22 \cdot 10^{-9}$. The low value indicates extremely low volatility and therefore we can also neglect the evaporation term (Bjerg et al., 2023). The mass balance for BAM will then look as follows:

$$\text{Accumulated} = \text{Net Transformed} - \text{Outflow}$$

Where $\text{Accumulated} = \frac{d(C_{BAM}(t) \cdot V_w)}{dt}$, and Net Transformed refers to the produced BAM from the degradation of DCB, which gives the following expression, $\text{Net Transformed} = k_1 \cdot C_{DCB}(t) \cdot V_w$, where C_{DCB} is the concentration of DCB. When C_{DCB} is multiplied by the rate constant, k_1, and the volume of the lake, V_w, the total expression Net Transformed describes the amount of BAM produced through the degradation of DCB. Lastly the we consider the outflow process, which can be expressed as $\text{Outflow} = U \cdot A_{out} \cdot C_{BAM}(t)$.

This results in the following differential equation for BAM:

$$\frac{d\big(C_{BAM}(t)\big)}{dt} = \frac{k_1 \cdot C_{DCB}(t) \cdot V_w - (U \cdot A_{out}) \cdot C_{BAM}(t)}{V_w}$$

3.1.2 Toxicity of DCB and BAM in the Lake

To be able to determine the extent of the toxicity levels of BAM, it is important to deduct the maximum concentration of BAM. This is done by solving the differential equation in an interval from 1 to 90 days. When extracting the maximum value we find $C_{BAM} = 0.02\, mg/L$ (see detailed calculation in Appendix 1, section 1C). As the value corresponds to $C_{BAM} = 20\, \mu g/L$ the concentration of BAM also exceeds the limit value for single pesticides in drinking water. The presence of pesticides in the water states that further filtration of the water is necessary before the water can be used as potable water. A plot of the concentration curve of DCB and BAM over time is seen in Figure 2.

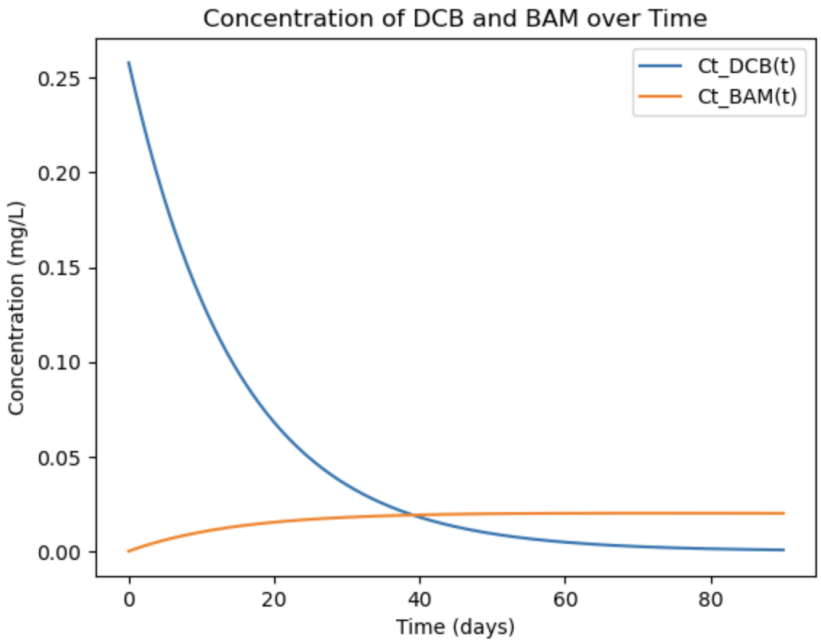


Figure 2: *Concentration of DCB and BAM over Time*

Based on the maximum determined concentration, it is relevant to assess whether this level exceeds toxicity thresholds for living fish in the lake. In this case it would be the concentration which is lethal to 50% of the population (LC_{50}) and the concentration which affects 50% of the population (EC_{50}). Extracted values and data appears in Table 2.

Table 2: *Comparison of LC_{50}/EC_{50} values and exceedance limits.*

Species	DCB LC_{50}/EC_{50} (mg/L)	BAM LC_{50}/EC_{50} (mg/L)	DCB Lim. Exceeded	BAM Lim. Exceeded
Rainbow trout	3.83	180	False	False
Fathead minnow	3.98	496	False	False
Duckweed	0.0279	75	True	False

According to values from the table, only the EC_{50} value for DCB for *Duckweed* is exceeded. Compared to LC_{50}, the exceeded EC_{50} is not as fatal for the organisms. We can conclude that the amount of DCB and BAM in the lake does not affect the ecosystem in a way where it is directly toxic and destructive for the fish population. Moreover it is only the DCB concentration that will affect the duckweed, but not lethally.

3.2 Pesticide Pollution at the Water Works

Now we consider a situation where the polluted water from Lake Gravel Pit is sent to an abstraction well belonging to the nearby water works (Figure 1, point 3.). The concentration of BAM in the water originating from the lake is assumed constant, equal to the maximum concentration in the lake to consider a worst case scenario. The water is diluted at the ab-

straction well, and the inlet concentration of BAM (BAM_{inlet}) is calculated, where we obtain a value of $BAM_{inlet} = 4 \cdot 10^{-4} mgL^{-1}$ (see Appendix 1, section 2A for calculations).

Afterwards the water is pumped to a waterwork, where it is treated by aeration and sand filtration (Figure 1, point 4.& 5.). Based on the existing sand filter, we determine the retardation factor to decide if BAM is retained by sorption. In determining the retardation factor, the $K_{d,BAM}$ value is calculated equivalent to previous calculation of the K_d value (see Appendix 1 section 2B). Although in this case we consider a new fraction of organic carbon $f_{oc} = 0.0011$. From Table 6.2 in the Compendium (Bjerg et al., 2023) the sorbent has a low carbon fraction of $f_{oc} < 1\%$, meaning new values for a & b are used. We deduct a value of $K_{d,BAM} = 1 \cdot 10^{-3}$. From calculations seen in Appendix 1, section 2B, we are able to compute a retardation value of $R = 1$. From the obtained value we can conclude BAM is not retained by the sand, and thereby follows the front line of the water moving through the filter.

To secure pesticide concentrations stay below danish legislative limits, an activated carbon filter is introduced (Figure 1, point 6.). A model for the dimensions of the cylindrical container has to be determined. Based on the geometrical properties of the cylinder given in Appendix 2, we are given a depth of the filter equal to $2m$. Analogue to:

$$\text{Geometry of cylinder} = \frac{height}{diameter} = 2\,m$$

Based on the equation, we have to use the volumetric properties of a cylinder. As the residence time, extracted from Appendix 2, is given in minutes, we have to convert the yearly output-flow in minutes. From where we get a value for the radius as $r = 0.97m$ (see Appendix 1, section 2C for calculations).

The dimensions of the height and diameter are computed, which gives the following values:

$$d = 1.937309\,\text{m} \qquad h = 3.874617\,\text{m}$$

The relationship between the height and diameter gives a ratio of 2, which verifies the obtained results of the dimensions.

3.2.1 Carbon Filter Life Time

At some point, the carbon filter needs to be changed due to functionality issues and lifespan for efficient filtration. First, its lifetime is determined neglecting dispersion. Based on the values given for the carbon filter in Appendix 2, we are able to determine the retardation factor, corresponding to a value of $R = 96601$. A large retardation value indicates the micropolutants and pesticides are retained by the filter. We calculate the seepage velocity

(calculations can be seen in Appendix 1, section 2D):

$$v_p = \frac{Q}{A\phi} = \frac{34.24 m^3 hours^{-1}}{2.947 m^2 \cdot 0.5} = 23.24\, m^3/hour$$

The time until the filter has to be changes can be calculated by the following equation:

$$t_s = \frac{h}{v_s} = \frac{Rh}{v_p} = 18.38 \text{ years}$$

The breakthrough curve without dispersion only accounting for advection appears from Figure 3 shown below.

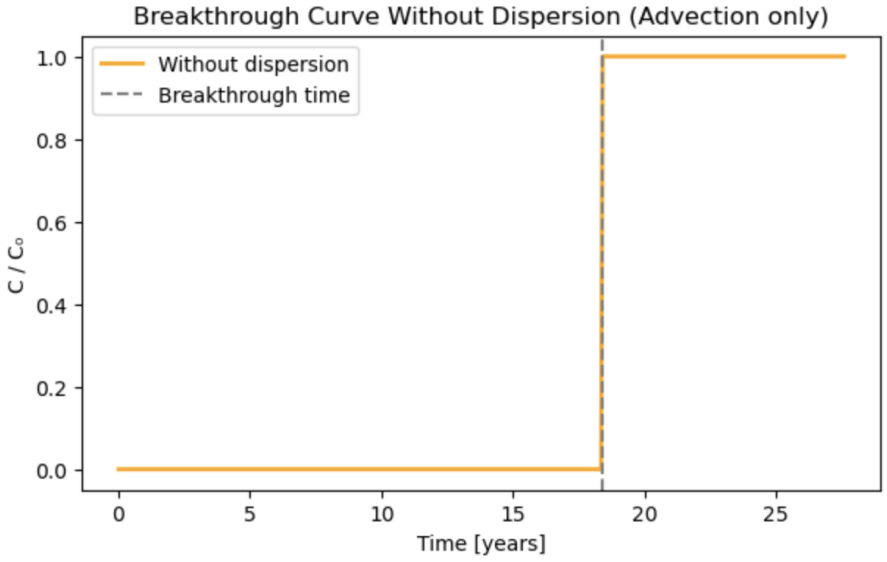


Figure 3: *Breakthrough Curve Without Dispersion (Advection only)*

In the simplified model without accounting for dispersion the lifetime of the filter equals 18.38 years (Appendix 1, section 2D). This value is obtained from the breakthrough time from the gray vertical dotted line on the curve. From the yellow line, we notice a constant concentration level until reaching the breakthough time with a sharp vertical spike front and then again reaching a constant level. However, this is a simplified model, only accounting for advection. To next simulate a more realistic application of the system, we establish a model that includes dispersion. The dispersion model is described by the following analytical solution:

$$F(x,t) = \frac{1}{2}\operatorname{erfc}\left(\frac{1-\theta_R}{2\sqrt{\frac{\theta_R}{\mathrm{Pe}}}}\right) + \frac{1}{2}\exp(\mathrm{Pe})\operatorname{erfc}\left(\frac{1+\theta_R}{2\sqrt{\frac{\theta_R}{\mathrm{Pe}}}}\right) \tag{4}$$

With $F(x,t)$ resembling the breakthrough function, describing the relative concentration in the column ($\frac{C_w}{C_0}$), Pe resembles the Péclet number $Pe = \frac{v_p h}{D}$, $\theta_R = \frac{\theta}{R}$ corresponding to the

retardation corrected time, and lastly $\theta = \frac{v_p t}{h}$ formulating the x-axis in a dimensionless time. The breakthrough curves for BAM in the carbon filter for both cases are shown in Figure 4.

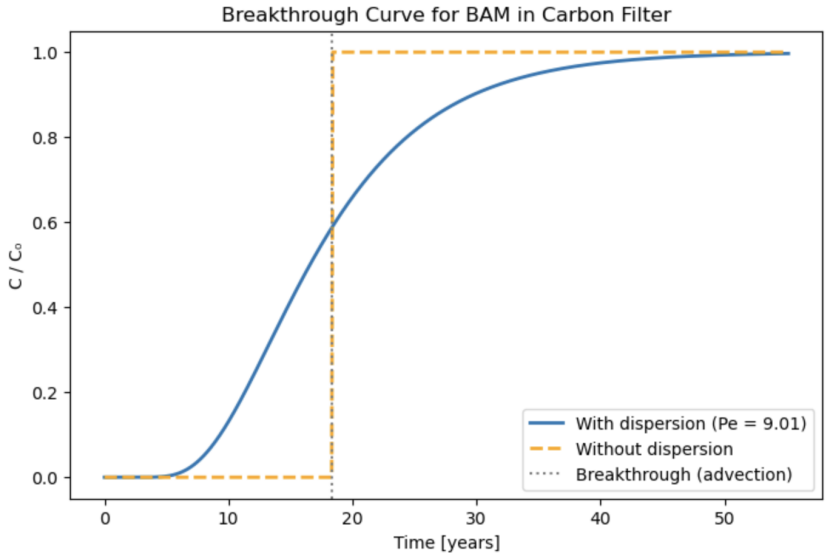


Figure 4: *Breakthrough Curves With and Without Dispersion*

From Figure 4 we see the breakthrough curve for BAM (blue line) with calculated Péclet number of 9.01 (see Appendix 1, section 2D). As the two functions describe the same situation, but with a variety in complexity we have to investigate whether the breakthrough time will be affected when accounting for dispersion. This is done by calculating a breakthrough ratio, which will be equal to the limit value $F(x,t) = \frac{C_{out}}{C_{BAM,max}} = 0.25$ (Appendix 1, section 2D for calculations). A new plot including the breakthrough curve and threshold for the BAM concentration in the activated carbon filter is shown in Figure 5.

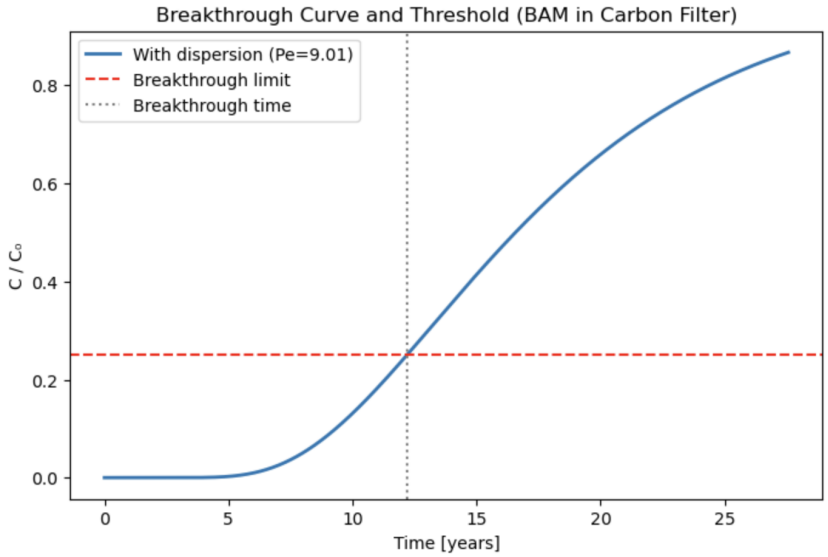


Figure 5: *Breakthrough Curve and Threshold*

The blue line resembles the dispersion function, the red horizontal dotted line, showing the breakthrough limit threshold, and the vertical grey line, showing the breakthrough time. As it appears from the model, it is clear that accounting for dispersion causes a displacement of the breakthrough time for the filter. The breakthrough time in this case is 12.22 years, corresponding to the time when the filter needs to be changed. This is equal to a depletion of lifetime in the filter of 6.15 years. For this reason it is important to account for dispersion, as otherwise problems will arise using a filter that has exceeded its lifetime.

3.2.2 Calcite Precipitation and Removal

When introducing the activated carbon filtration (Figure 1, point 6), it can lead to clogging if iron and manganese oxides precipitate in the filter. These oxides therefore need to be removed beforehand, which requires replacing the cascade aerator with an aeration tank (Figure 1, point 4). Initially, the water contains CO_2 far above equilibrium, and aeration is used to reduce this level to nine times the atmospheric equilibrium concentration.

To evaluate the potential for calcite ($CaCO_3$) precipitation, the dissolution equilibrium is considered:

$$\mathrm{CaCO_3} \rightleftharpoons \mathrm{Ca^{2+}} + \mathrm{CO_3^{2-}}$$

The analysis is based on the solubility product $K_{\mathrm{sp}} = 10^{-8.36}$ at 10°C, a calcium concentration of 94 mg/L (Appendix 2), a CO_2 concentration corresponding to 9 times the atmospheric equilibrium level ($9 \cdot 420$ ppm), and an assumed activity coefficient of $\gamma = 1$.

Using the carbonate system relationships, the concentration of $\mathrm{CO_3^{2-}}$ is first determined. This calculated concentration is then combined with the measured Ca^{2+} concentration to evaluate the ion product defined as $[Ca^{2+}] \cdot [CO_3^{2-}]$. The ion product is calculated to be $3.82 \cdot 10^{-8}$, and since this is larger than the solubility product it indicates a precipitation of $CaCO_3$.

However, only ideal systems will have an activity factor of $\gamma = 1$, since we are examining groundwater, the ions will interact and an activity factor less than 1 should be applied. Since the conductivity is stated to be 61 mS/m in Appendix 2, activity factors $\gamma_1 = 0.95$ and $\gamma_2 = 0.75$ are applied, in reference to Table 1.1 from the Compendium(Bjerg et al., 2023). γ_1 applies to ions with a charge of ± 1 and γ_2 applies to ions with charge ± 2. Once again, the concentration of $\mathrm{CO_3^{2-}}$ is determined and the ion product $\gamma_2[Ca^{2+}] \cdot \gamma_2[CO_3^{2-}]$ is calculated. This time the ion product is calculated to be $3.015 \cdot 10^{-8}$, which is still greater than the solubility product, and therefore precipitation of calcite still occurs.

3.2.3 Iron Precipitation and Removal

To increase the lifetime of the activated carbon filter and be below required limit of iron in potable water, iron precipitation must also be accounted for in the water treatment. This is handled in the aeration process of the water treatment where ferrous iron (Fe^{2+}) is oxidized to ferric iron (Fe^{3+}) which then precipitates as ferric hydroxide $Fe(OH)_3$ according to the reaction below.

$$4Fe^{2+} + O_2 + 10H_2O \leftrightarrow 4Fe(OH)_3(s) + 8H^+ \tag{5}$$

To ensure optimal precipitation of ferric hydroxide the efficiency of different reactor setups are calculated. For this application the efficiency of reactor designs for CMFR, CMFR in series and PFR are calculated. From Appendix 2, the defined values are a first order reaction rate, iron concentration in intake water and residence time. With this information the output concentration of iron for the reactor designs may be derived (see Appendix 1, section 2G). Comparing this to the maximum acceptable iron concentration in potable water of 200 $\mu g/L$ we get the following overview.

Table 3: *Output iron concentration for different aeration reactor designs*

Reactor Design	Output Iron Conc.	Unit	Below Conc. Limit
CMFR	428.57	µg/L	False
CMFR in Series	30.61	µg/L	True
PFR	0.014	µg/L	True

We see from table 3 that both CMFR in series and PFR are suitable reactor design to achieve acceptable iron concentration in the potable water.

3.3 Potential of Alternative Water Well

As a solution to avoid the BAM pollution in the water work, a new groundwater well is considered as a replacement for the initial well. To establish whether this is a good solution or not, a few analyses are made using data on the raw water from the well. Below Table 4 presents the concentrations of cations and anions in the raw water.

Table 4: *Cation and anion concentrations from alternative water well.*

Cations	**[mg/L]**	**Anions**	**[mg/L]**
Sodium, Na^+	50	Chloride, Cl^-	200
Potassium, K^+	15	Nitrate, NO_3^-	2.6
Calcium, Ca^{2+}	210	Hydrogen carbonate, HCO_3^-	385
Magnesium, Mg^{2+}	37	Sulfate, SO_4^{2-}	205
Iron, Fe^{2+}	2		
Ammonium, NH_4^+	1		

The given water analysis is complete, since the electro neutrality deviation is -0.0016%, and any deviation below $\pm 5\%$ is accepted. This means no major ions are missing and data is sufficient (see Appendix 1, section 3A for calculations).
The hardness of the raw water is also calculated. From the list of present ions in Table 4, only Mg^{2+} and Ca^{2+} contribute to the hardness. The concentrations $[mg/L]$ for the two ions are converted to molarity $[mol/L]$ by multiplying the concentrations by their respective molar masses $[mg/mol]$. The hardness $[meq/L]$ for each ion is then calculated by multiplying each of their molarity by their charge, for both Mg^{2+} and Ca^{2+} it would be 2. The hardness can then be converted from meq/L to the unit °dH by multiplying with a conversion factor of 2.8. The total hardness of the water in the considered water well can be found by adding the hardness of each of the contributors, Mg^{2+} and Ca^{2+}. Below in Table 5, the hardness presented in different units, while calculations can be found in Appendix 1, sections 3A and 3B.

Table 5: *Water hardness from calcium and magnesium concentrations.*

Hardness	**meq/L**	**°dH**
Ca^{2+}	10.48	29.38
Mg^{2+}	3.04	8.54
Total hardness	**13.52**	**37.92**

The total hardness is calculated to be 37.92 $°dH$ (see Appendix 1, section 3B), which is considered "extremely hard", according to the map of the hardness of water in Denmark, where any hardness levels $> 30\ °dH$ is "extremely hard" (Baun, 2025). The hardness does not have a specific limit value, but a high hardness can be very costly from an economic perspective. A high hardness would result in scaling in machines, which would then decrease the efficiency, and increase maintenance of appliances and machines used in all industries and areas of society. However, the hardness can be decreased by adding a softening step at the water works. This could be by adding a pellet reactor, which would result in the precipitation of $CaCO_3$ and thereby remove Ca^{2+} from the water (Tretow-Loof et al., 2025). Since

Ca^{2+} is the largest contributor to the total hardness, it would effectively soften the water.

Evaluating the values of other compounds in the raw water from the considered well, is also needed, to see if any limit values for drinking water are exceeded. Below in Table 6, the concentrations in the water at the well as well as limit values for drinking water in Denmark are displayed.

Table 6: *Comparison of measured concentrations with limit values.*

	$\mathbf{NO_3^-}$	$\mathbf{NH_4^+}$	$\mathbf{Cl^-}$	$\mathbf{Fe^{2+}}$	$\mathbf{SO_4^{2-}}$	$\mathbf{Na^{2+}}$
Limit value [mg/L] *	50	0.05	250	0.20 †	250	175
Measured concentration [mg/L]	2.6	**1**	200	**2.0**	205	50

Note: * Limit values are taken from the Danish drinking water regulation ("Bekendtgørelse om vandkvalitet og tilsyn med vandforsyningsanlæg", 2023).
† Limit value is for Fe total, while the measured concentration is for Fe^{2+}.

Table 6 illuminates exceeded limit values for both ammonium and iron in the raw water from the alternative water well. This means it would need treatment before the well could be used as a supply for drinking water. The Fe^{2+} concentration exceeds the limit value for total Fe, which makes treatment necessary. This could be done is by adding an aeration step, which would cause both Fe^{2+} and Fe^{3+} to precipitate as iron hydroxide, as described earlier. To lower the ammonium value a nitrification step could be added, where a group of microorganisms convert ammonium into nitrite, and then from nitrite to nitrate (Bjerg et al., 2023). Since the measured nitrate concentration is much lower than the limit value, the conversion from ammonium to nitrate, the solution would make the compound concentrations stay within the limit values on all the parameters for drinking water in Denmark.

3.4 Overall discussion

Initially the concentrations for BAM and DCB in the lake, equal to values of respectively $BAM = 20\mu g/L$ and $DCB = 258\mu g/L$, highly exceeds threshold limits in regards to Danish legislation for pesticide concentrations allowed in drinking water, 0.1 $\mu g/L$ for single pesticides and 0.5 $\mu g/L$ in total. Although at this stage the water has not undergone any treatment, and is not intended to be used for drinking water yet. Previous results obtained for EC_{50} and LC_{50}, indicates that initial concentrations for both BAM and DCB in the lake are not high enough to cause any fatal effects on the ecosystem in the lake, and does not cause any critical destructive toxicity. When reaching the groundwater well, the concentration based on the maximum concentration of BAM decreases to a much lower value of $C_{BAM} = 0.4\mu g/L$, however it is still 4 times beyond the regulatory limit. Therefore additional treatment is needed to meet the requirements of drinking water standards. Since the

sand filter does not retain the pesticides in the water, reflected by the retardation factor with a value of $R = 1$, establishing further filtration through an activated carbon filter sis necessary. The corresponding retardation factor for the carbon filter with a value of $R = 966001$ directly indicates that the micro-pollutants are retained in the filter. As it appears from the breakthrough curves, advection alone does not provide the necessary results, as when accounting for dispersion, the breakthrough time decreases from 18.38 years to 12.22 years. This demonstrates that the dispersion affects the total transport time, which is decreased for the time until the polluted molecules passes through the filter.

Yet introducing this filter brings secondary challenges, as the filter leads to clogging of iron and manganese oxides that precipitate in the filter. This means the aeration process has to be reconsidered. As we see an oversaturated solution from the ion product, an iron oxidation step is added, where the iron after the reaction will precipitate as solid ferric hydroxide. To discover which reactor model is the best fit, different models were evaluated in regards to the maximum allowed iron concentration limit in drinking water of 200 $\mu g/L$. A PFR reactor is, in this case, the most suitable- and efficient choice of model, since it has the lowest outlet concentration in the worst case scenario of $C_{out,PFR} = 14 \mu g/L$. Despite that a CMFR-reactor in series is also a possible choice with an outlet concentration of $C_{out,CMFRseries} = 30.61 \mu g/L$, which is also below the limit value.

To completely escape the issue of the pesticide pollution, an alternative water well is considered. When analyzing the quality of the raw water from the area of the new well, the concentrations of iron and ammonium exceed limit values, which means it would require further treatment to meet the standards for drinking water in Denmark. The hardness of the water is also examined, and is found to be categorized as extremely high. Even though there are no limit values for hardness the water would most likely need softening treatment because of the economic issue related to maintenance of machines, which is caused by the scaling. The suggested treatments for the water well includes a pellet reactor to soften the water, aeration to remove iron, as well as a nitrification process, which would lower levels of ammonium. This means the considered alternative requires more processes in the treatment of the water, before it meets the standard for danish drinking water, but it completely escapes the issue of pesticide pollution.

4 Conclusion

The aim of this report is to provide an overview of the effects from point source pollution by a pesticide with batch discharge into a lake with a nearby drinking water abstraction well. The overview includes analysis of toxicology and necessary water treatment processes to meet acceptable danish regulatory standards for potable water. The pesticide in focus in this report, is DCB and its degradation product BAM.

Assumptions were made throughout the report for the simplification of calculations. To maintain reliability of results a worst-case-scenario outlook was enforced across calculations to induce a safety margin for uncertainties.

Assuming linear sorption and the lake acting as a CMFR reactor, the concentration of BAM peaks at 20 $\mu g/L$ and occurs 72.6 days after the 10 kg of DCB is released. Based on provided EC_{50} and LE_{50} values, there are no critical toxicological effects on organisms in the lake.

Assuming constant peak BAM concentration in the lake, the BAM concentration in the inlet water to the water works will be 0.4 $\mu g/L$, 4 times the regulatory limit for a single pesticide compound. We conclude water treatment is necessary.

Calculations for sand filtration showed no retardation effect for BAM and is insufficient. Activated carbon filtration showed much greater retardation of the pollutant. Finding the breakthrough ratio at 0.25 and including dispersion, the lifetime for a carbon filter with required size dimensions is derived to 12.2 years.

Due to the concentration of CO_2 being well above atmospheric equilibrium in the carbon filter, precipitation of calcium carbonate will occur, when accounting for the activity coefficients ≤ 1. Additionally, iron levels in the inlet water is above legislative limits. We conclude an aeration process is necessary. Comparing iron precipitation efficiency of different reactor designs; CMFR-in-series and PFR designs achieved acceptable iron concentrations at 30.61 $\mu g/L$ and 0.014 $\mu g/L$ respectively.

The alternative groundwater well showed sufficient data for complete water analysis. From given data, the water hardness is derived to an extraordinarily high 37.87 $^{\circ}dH$. A water softening system should be taken into consideration before being chosen as an alternative drinking water source, as well as treatment processes to handle levels of iron and ammonium, which exceed their respective limit values for potable water.

5 Literature

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6 Appendix

Appendix 1

A separate Jupyter notebook file containing raw data treatment has been uploaded as additional material for this report. The material consists of calculations excluded from this report, due to length limitations and practicality.

Appendix 2

The below information is given as additional information for "The Grand Exercise" (Kaarsholm et al., 2024), under the section "Substance properties and technical data". The values and information are used throughout the report for calculations.

Pesticides

Compound	**CAS**	K_H (atm·m^3·mol^{-1})	**Toxicity data**		
			LC_{50} (mg/L)	EC_{50} (mg/L)	
			Rainbow trout	Fathead minnow	Duckweed
DCB	1194-65-6	$8.3 \cdot 10^{-6}$	3.83	3.98	0.0279
BAM	2008-58-4	$1.22 \cdot 10^{-9}$	180	496	75

LC_{50} = concentration lethal to 50% of the population. EC_{50} = concentration affecting 50% of the population. Data from ECOTOX database (2004) and Rippen (2003), *Handbuch Umweltchemikalien*.

Dichlobenil, DCB
Half-life in lake: $t_{1/2} = 128$ d (first-order kinetics)
Log $K_{ow} = 2.70$
BAM
K_d (active carbon) $= 1.15\,m^3/g$ active carbon
Log $K_{ow} = 0.77$

Lake Gravel Pit

Sediment property	**Value**	**Unit**
Bulk density, ρ_b	1600	kg/m^3

The Waterworks

Filter material (existing sand filter)	Value	Unit
Filter depth	2.00	m
Bulk density (packet)	1300	kg/m^3
Porosity, ϕ	0.50	–
Organic carbon	0.0011	%

Active carbon filter

Parameter	Value	Unit
Residence time (empty bed), T_h	20	min
Geometry: height:diameter ratio	2	–
Porosity, ϕ	0.50	–
Bulk density, ρ_b	0.42	kg/L
Dispersion coefficient for pesticides, ε	10	m^2/h

Existing plant raw water analysis

Parameter	Value	Unit
Temperature	10	°C
Oxygen	0	mg/L
pH	7.5	–
TAL	4.61	meq/L
Conductivity	61	mS/m
Calcium	94	mg/L
Iron	6	mg/L